

Dealing with Sensitive Data



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Module Overview



What are sensitive data?

Handling sensitive data

- Inputs and outputs
- State data

Implement AWS Secrets Manager

Remove last provisioner





Defining Sensitive Data



**Sensitive data are
information that should be
kept private and secured
via access controls.**



Sensitive Data



Types of data

- API keys, passwords, and credential tokens
- IP addresses, host names, database ports

Sensitivity is context specific

- Consult compliance and security teams



Sensitive Argument

main.tf

```
variable "api_key" {  
  type      = string  
  description = "API Key for application"  
  sensitive  = true  
}  
  
output "private_key" {  
  description = "SSH Key for connections"  
  sensitive    = true  
}
```

> terraform output

```
private_key = (sensitive)
```



Sensitive argument

terraform.tfstate

```
{
  "version": 4,
  "terraform_version": "1.5.0",
  "serial": 1,
  "lineage": "c96cd949-cf0f-14b6-e336-6b87164c696b",
  "outputs": {
    "secret_value": {
      "value": "$uper$ecret@PIK3y",
      "type": "string",
      "sensitive": true
    }
  },
  "resources": [],
  "check_results": null
}
```



Sensitive Data in State



State data contents

- Resources, data sources, and outputs
- Attributes stored in plaintext

Securing state

- Encryption at rest and in transit
- Robust access control scheme

Store sensitive data outside of Terraform

Current Configuration

main.tf

```
variable "api_key" {  
  type      = string  
  description = "(Required) API key for web app..."  
}  
  
resource "terraform_data" "webapp" {  
  ...  
  provisioner "file" {  
    ...  
    api_key = var.api_key  
    ...  
  }  
}
```



Current Configuration

main.tf

```
variable "api_key" {  
  type          = string  
  description = "(Required) API key for web app..."  
  sensitive     = true  
}  
  
resource "terraform_data" "webapp" {  
  ...  
  provisioner "file" {  
    ...  
    api_key = var.api_key  
    ...  
  }  
}
```



Data Source Option

main.tf

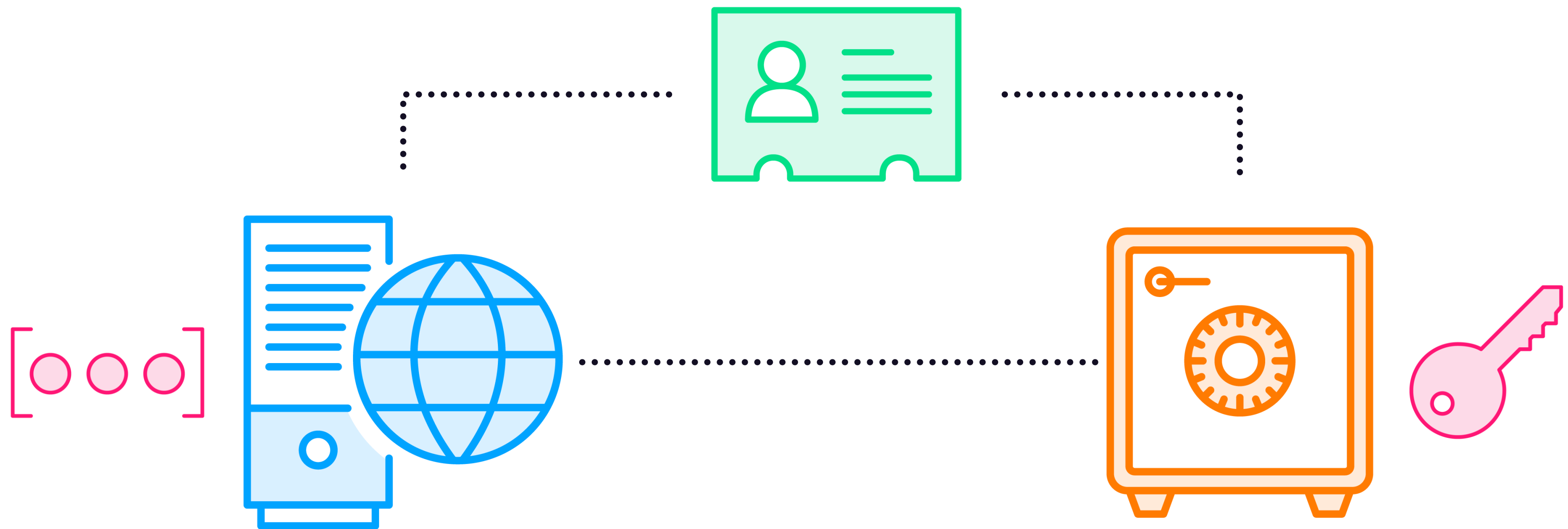
```
data "aws_secretsmanager_secret_version" "api_key" {  
    secret_id = var.secret_id  
}
```

In terraform.tfstate

```
...  
{  
    "mode": "data",  
    "type": "aws_secretsmanager_secret_version",  
    "name": "api_key",  
    "instances": [  
        {  
            "attributes": {  
                "secret_string": "$uper$ecret@PIK3y"  
            }  
        }  
    ]  
}  
...
```



Storing Secrets Outside of Terraform



Demo



Provision the secret and role

Create and assign instance profile

Confirm secret access



Demo



Add Systems Manager Parameter Store entries

Grant access through instance profile

Remove last provisioner



Module Summary



Handling sensitive data with Terraform

- Sensitive argument
- Terraform state

Keeping secrets out of Terraform state

Store application config in data sources

What's Next?



Terraform Cloud

Terraform Certified Associate

- Certification guide

HashiCorp family

- Vault, Consul, Nomad

DevOps tools

- SCM, CI/CD, containers, config management



Course Summary



Terraform is part of a larger ecosystem

You are part of a large ecosystem

Learn tools, focus on concepts

Have fun!



Go Build Something Great!



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